Project Report

ON

STATISTICAL TECHNIQUES USING R LAB

TITLE

RetaillQ

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the mini project report entitled **"RetailIQ: Retail Intelligence & Business Performance Analytics System"** is a bona fide work carried out by **Anjali Kumari,** (**UID** : **26MCD10064)**, a student of Master of Computer Applications (MCA) at Chandigarh University, under my guidance and supervision.

The project has been completed as a part of the academic requirements for the R Programming Lab. The work presented in this report is based on the student's efforts and demonstrates the practical application of R programming, data analysis, and data visualization techniques.

To the best of my knowledge, this work has not been submitted previously, either in part or in full, for the award of any other degree or academic qualification.

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I would like to express my sincere gratitude to everyone who supported and guided me in completing my mini project, “RetailIQ: Retail Intelligence & Business Performance Analytics System.”

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This project helped me gain practical knowledge of R programming, data analysis, dashboard development, and graphical representation of retail data. Finally, I am grateful to my family and friends for their constant encouragement and support.

Thank you to everyone who contributed directly or indirectly to the successful completion of this project.

I am also thankful to my classmates and friends for their helpful suggestions and cooperation during the completion of this project. This project has been a valuable learning experience that enhanced my technical skills and helped me understand the practical use of R programming in data analysis.

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ABSTRACT

Retail businesses generate a large amount of data related to sales, products, customers, and profits. Analyzing this data is important for understanding business performance and making better decisions. The project “RetailIQ: Retail Intelligence & Business Performance Analytics System” is developed using R programming to analyze and visualize retail sales data through an interactive dashboard.

The system uses R libraries such as Shiny, shinydashboard, ggplot2, dplyr, and DT to provide an interactive and user-friendly interface. It presents important business indicators such as total revenue, profit, number of orders, and profit margin. The dashboard also includes graphical representations such as monthly sales trends, product-wise revenue, region-wise sales, category-wise revenue distribution, and profit analysis.

The project allows users to apply filters based on date, product category, and region to explore the data more effectively. Different graphs and visualizations help identify sales patterns, compare product performance, and understand business profitability.

The main aim of this project is to demonstrate the practical use of R programming in data analysis and visualization. It provides a simple and interactive way to understand retail business data and supports better decision-making through graphical analysis.

INTRODUCTION

Retail business is one of the important sectors of the economy. It involves the sale of products directly to customers through stores and online platforms. Every retail business generates data related to product sales, quantities, revenue, costs, and profits. Proper analysis of this data helps businesses understand their performance and make effective decisions.

RetailIQ: Retail Intelligence & Business Performance Analytics System is a mini project developed using R programming. The main purpose of this project is to analyze retail sales data and present meaningful information through an interactive dashboard. The system provides a graphical representation of sales and profit-related information, making the data easier to understand.

The project uses R programming along with libraries such as Shiny, shinydashboard, ggplot2, dplyr, and DT. These libraries help in developing an interactive user interface, performing data analysis, creating graphs, and displaying records in a tabular format.

The dashboard includes important performance indicators such as total revenue, total profit, number of orders, and profit margin. It also provides different visualizations, including monthly sales trends, product-wise revenue, region-wise sales, and profit analysis. Users can apply filters based on date, category, and region to explore specific portions of the dataset.

This project demonstrates how R programming can be used to transform raw retail data into useful business insights. It also provides practical experience in data manipulation, statistical analysis, graphical visualization, and interactive dashboard development.

2. PROBLEM STATEMENT:

Retail businesses deal with a large amount of sales data, including product details, sales quantity, revenue, cost, and profit. Managing and analyzing this data manually can be timeconsuming and may make it difficult to identify important business trends.

Without proper visualization, it becomes challenging to understand which products generate the highest revenue, which regions perform better, and how sales change over time. Businesses may also face difficulties in comparing revenue and profit or identifying variations in product performance.

Therefore, there is a need for an interactive system that can analyze retail sales data and present the results in a simple and understandable format.

The RetailIQ: Retail Intelligence & Business Performance Analytics System aims to solve this problem by using R programming to analyze retail data and display meaningful insights through an interactive dashboard. It provides graphical analysis, key performance indicators, and filtering options to help users explore sales performance efficiently.

3. OBJECTIVES OF THE PROJECT:

The main objective of the project "RetailIQ: Retail Intelligence & Business Performance Analytics System" is to analyze retail sales data using R programming and present meaningful business insights through an interactive dashboard.

The specific objectives of the project are:

1. To analyze retail sales data and understand overall business performance.
2. To calculate important performance indicators such as total revenue, total profit, number of orders, and profit margin.
3. To visualize sales trends using different graphs and charts.
4. To compare product-wise revenue and profit.
5. To analyze sales performance across different regions and product categories.
6. To provide interactive filters based on date, category, and region.
7. To identify sales patterns and understand changes in business performance.
8. To develop an attractive and user-friendly dashboard using R programming.
9. To gain practical knowledge of data manipulation, analysis, and visualization using R libraries.
10. To demonstrate how data visualization can support better business decision-making.

4. SCOPE OF THE PROJECT:

The scope of the project "RetailIQ: Retail Intelligence & Business Performance Analytics System" is to provide an interactive platform for analyzing and visualizing retail sales data using R programming. The system focuses on understanding business performance through graphical representation and statistical analysis.

The project includes the following areas:

1. Sales Analysis: Analyzes sales data to calculate total revenue, profit, and other important business indicators.
2. Product Performance: Helps identify products that generate higher revenue and profit.
3. Regional Analysis: Compares sales performance across different geographical regions.
4. Category Analysis: Displays revenue distribution among different product categories.
5. Trend Analysis: Shows changes in sales and profit over time through graphical representations.
6. Interactive Dashboard: Provides filters that allow users to explore data according to date, category, and region.
7. Data Visualization: Uses graphs and charts to present complex data in a simple and understandable form.

The system is developed using a sample retail dataset and can be used for educational purposes to understand the practical application of R programming in business analytics.

In the future, the project can be extended by integrating real-time sales data, customer analysis, sales forecasting, and advanced machine learning techniques.

. HARDWARE AND SOFTWARE REQUIREMENTS:

The development of the RetailIQ: Retail Intelligence & Business Performance Analytics System requires basic hardware and software resources. These resources provide the necessary environment for developing, executing, and testing the application.

5.1 Hardware Requirements:

Hardware Component	Minimum Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB
Hard Disk	500 GB HDD or 256 GB SSD
Display	Standard Monitor
Input Devices	Keyboard and Mouse

5.2 Software Requirements:

Software	Description
Operating System	Windows 10/11 or equivalent
Programming Language	R
IDE	RStudio
Dashboard Framework	Shiny
UI Framework	shinydashboard
Data Visualization	ggplot2

Data Manipulation	dplyr

5.3 Software Description:

- R: Used for statistical computing, data analysis, and programming.
- RStudio: Provides an integrated development environment for writing and executing R code.
- Shiny: Used to develop interactive web applications using R.
- shinydashboard: Helps create a structured dashboard interface with navigation and information panels.
- ggplot2: Used to create graphs and charts for visualizing retail data.
- dplyr: Used for data filtering, grouping, summarizing, and manipulation.

DT: Used to display interactive data tables with searching, sorting, and pagination features.

6. TOOLS AND TECHNOLOGIES USED:

The RetailIQ project is developed using R programming and various supporting libraries. These tools help in data processing, analysis, visualization, and dashboard development.

6.1 R Programming Language:

R is a programming language used for statistical computing, data analysis, and graphical representation. In this project, R is used to process retail sales data, calculate revenue and profit, and generate analytical results.

6.2 RStudio:

RStudio is an integrated development environment (IDE) for R. It provides an environment for writing, executing, debugging, and managing R programs. The complete project is developed and executed in RStudio.

6.3 Shiny:

Shiny is an R package used to create interactive web applications. It allows users to interact with the dashboard through filters and controls without requiring advanced programming knowledge.

6.4 shinydashboard:

The shinydashboard package is used to create a structured dashboard layout. It provides features such as a dashboard header, sidebar navigation, and content panels.

6.5 ggplot2:

The ggplot2 package is used to create different types of graphs and charts. In this project, it is used for sales trends, product comparisons, profit analysis, and other visualizations.

6.6 dplyr:

The dplyr package is used for data manipulation and transformation. It helps in filtering records, grouping data, and calculating summarized values.

6.7 DT

The DT package is used to display retail data in an interactive table. It provides features such as searching, sorting, and pagination, making it easier to explore the dataset.

6.8 Technologies Summary

Technology	Purpose
R	Data analysis and calculations
RStudio	Development environment
Shiny	Interactive application development
shinydashboard	Dashboard layout and navigation
ggplot2	Graphical visualization
dplyr	Data manipulation
DT	Interactive data table

Next section: Dataset Description.

7. DATASET DESCRIPTION:

The RetailIQ project uses a sample retail sales dataset to analyze business performance. The dataset contains information about products, sales quantity, revenue, cost, profit, and different regions.

The dataset is generated using R programming and contains 100 records. It is used to demonstrate the working of the dashboard and its analytical features.

7.1 Dataset Attributes

Attribute	Description
Date	Date of the sales transaction
Product	Name of the product
Category	Product category
Quantity	Number of units sold
Region	Region where the sale occurred
Sales	Total revenue generated
Cost	Total cost of products sold
Profit	Difference between sales and cost
Profit Margin	Percentage of profit earned

7.2 Products Used:

The dataset includes the following products:

- Laptop
- Mobile
- Headphones
- Tablet
- Keyboard
- Mouse

The data is used to perform product-wise, category-wise, region-wise, and monthly sales analysis.

Screenshot placement: Yahan dashboard ke Data Records tab ka screenshot add karna.

Figure 1: Retail Sales Dataset Displayed in the Dashboard

8. SYSTEM ANALYSIS AND METHODOLOGY:

8.1 System Analysis:

System analysis is the process of understanding the requirements and functionality of the proposed system. The RetailIQ project is designed to analyze retail sales data and present useful business information through an interactive dashboard.

The system processes sales records and calculates important performance indicators such as total revenue, total profit, number of orders, and profit margin. It also provides graphical analysis to understand product performance, regional sales, and monthly trends.

The dashboard allows users to apply filters based on date, product category, and region. This helps users explore the dataset and understand business performance more effectively.

8.2 Methodology:

The project follows a simple data analysis methodology consisting of the following steps:

1. Data Generation: A sample retail dataset containing 100 records is generated using R.
2. Data Processing: The dataset is processed using the `dplyr` package.
3. Calculation: Revenue, cost, profit, and profit margin are calculated.
4. Data Filtering: Users can filter data by date, category, and region.
5. Data Visualization: Graphs are created using `ggplot2`.
6. Dashboard Development: An interactive dashboard is developed using Shiny and shinydashboard.
7. Output Analysis: The results are displayed through KPI cards, graphs, and an interactive data table.

9. SYSTEM DESIGN AND ARCHITECTURE:

9.1 System Architecture:

The system architecture of RetailIQ represents the overall structure of the application. It explains how the retail dataset is processed and displayed through an interactive dashboard.

The system consists of three main layers:

1. Data Layer: Contains the retail sales dataset, including product, date, quantity, region, sales, cost, and profit.
2. Processing Layer: Uses R and dplyr to process the data, calculate business metrics, and apply user-selected filters.
3. Visualization Layer: Uses Shiny, shinydashboard, and ggplot2 to display KPI cards, graphs, and interactive tables.

10. IMPLEMENTATION:

The RetailIQ project is implemented using R programming in RStudio. The application is developed using the Shiny framework to provide an interactive dashboard for retail sales analysis.

The implementation involves generating a sample retail dataset, processing the data, calculating business metrics, and displaying the results through graphs and tables.

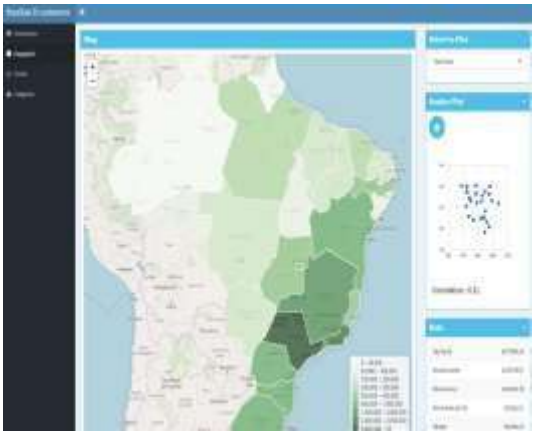
The project uses different R libraries to perform data manipulation, visualization, and dashboard development.

10.1 Implementation Steps:

1. Import the required R libraries.
2. Generate the sample retail sales dataset.
3. Calculate sales, cost, profit, and profit margin.
4. Create the dashboard layout using Shiny and shinydashboard.
5. Add filters for date, category, and region.
6. Generate graphs using ggplot2.
7. Display the dataset using the DT package.
8. Run the application in RStudio.

10.2 Screenshot Placement:

```r
library(shiny)
ui <- fluidPage(
    mainPanel(
        sliderInput("obs",
            "Number of observations",
            min = 1,
            max = 5000,
            value = 100),
        plotOutput("distPlot")
    )
)
server <- function(input, output) {
    output$distPlot <- renderPlot({
        dist <- rnorm(input$obs)
        hist(dist,
            col="purple",
            xlab="Random values")
    })
}
shinyApp(ui = ui, server = server)
```



11.2 Monthly Sales Trend:

The monthly sales trend graph displays the changes in revenue over different months. It helps understand the sales pattern and identify months with higher or lower revenue.



Figure 6: Monthly Sales Trend

11.3 Quantity vs Sales Analysis:

The scatter plot represents the relationship between the quantity of products sold and the total sales amount. It helps understand how sales revenue changes with the quantity sold.



11.4 Product-wise Profit Analysis:

The product-wise profit graph compares the profit generated by different products. It helps identify which products contribute more towards business profitability.



Figure 8: Product-wise Profit Analysis

12. RESULTS AND OUTPUT:

The RetailIQ: Retail Intelligence & Business Performance Analytics System was successfully developed using R programming and the Shiny framework. The system provides an interactive dashboard for analyzing retail sales data.

The dashboard displays key performance indicators such as total revenue, total profit, number of orders, and profit margin. It also provides graphical visualizations to analyze monthly sales trends, product-wise revenue, regional sales, category-wise revenue distribution, and profit variation.

The interactive filters allow users to explore the dataset based on date, category, and region. The Data Records section provides an interactive table for viewing and analyzing sales records.

The results demonstrate that R programming can be effectively used to process retail data, perform calculations, and present meaningful business insights through graphical visualization.

12.1 Final Result:

The project successfully achieves its objective of developing an interactive retail analytics dashboard. It provides a simple and effective way to understand sales performance, compare products, analyze regional revenue, and evaluate business profitability.

The project also helped in gaining practical knowledge of R programming, data manipulation, graphical visualization, and Shiny dashboard development.

13.1 Advantages:

The RetailIQ: Retail Intelligence & Business Performance Analytics System provides several advantages for retail data analysis.

1. Interactive Dashboard: Provides a user-friendly interface for exploring retail sales data.
2. Easy Data Visualization: Presents complex data through graphs and charts, making it easier to understand.
3. Performance Analysis: Displays important metrics such as revenue, profit, orders, and profit margin.
4. Product Comparison: Helps compare the revenue and profit generated by different products.
5. Regional Analysis: Provides insights into sales performance across different regions.
6. Data Filtering: Allows users to filter data by date, category, and region.
7. Time Saving: Reduces the effort required for manually analyzing sales records.
8. Learning Opportunity: Provides practical experience in R programming, data manipulation, and dashboard development.

13.2 Limitations:

Although the RetailIQ system provides useful analytical features, it has certain limitations.

1. Sample Dataset: The project uses a sample dataset and may not represent actual retail business conditions.
2. Limited Data Size: The current system contains 100 records, which may not be sufficient for large-scale business analysis.

3. No Real-Time Data: The dashboard does not automatically update using live retail transactions.

4. Limited Prediction: The system focuses on data analysis and visualization rather than advanced sales forecasting.
5. Basic Security: The application does not include user authentication or advanced data security features.
6. Dependency on Data Quality: Incorrect or incomplete data may affect the accuracy of the analysis.

14. LEARNING OUTCOMES:

Through this project, the following learning outcomes were achieved:

1. Gained practical knowledge of R programming.
2. Learned to process and analyze retail sales data.
3. Understood the calculation of revenue, cost, and profit.
4. Learned to create graphs using the `ggplot2` package.
5. Gained experience in developing interactive dashboards using Shiny.
6. Learned to display data using interactive tables.
7. Improved understanding of data visualization and business analysis.

15. CONCLUSION:

The RetailIQ: Retail Intelligence & Business Performance Analytics System was successfully developed using R programming and the Shiny framework. The project provides an interactive dashboard to analyze retail sales data through graphs, charts, and key performance indicators.

This system helps users understand product performance, sales trends, regional revenue, and business profitability. It also provides filtering options for better data exploration.

Overall, the project demonstrates the practical application of R programming in data analysis and visualization. It helped in gaining valuable knowledge of data processing, graphical representation, and interactive dashboard development.

16. GITHUB REPOSITORY:

The source code of the project “RetailIQ: Retail Intelligence & Business Performance Analytics System” has been uploaded to GitHub. The repository contains the R source code and project-related files.

The GitHub repository can be accessed using the following link:

GitHub Repository Link:
[Paste Your GitHub Repository Link Here]